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## **COMPUTER NETWORKS**

Complete Course Notes

By ***Muhammad Usman Ghani (Aspire College Vehari Campus)***

***(Credit Hours: 3 | Lectures: 2, Labs: 1 | Prerequisites: None)***

1. Introduction to Computer Networks, 4th Ed. — A. S. Tanenbaum, Prentice Hall, 2003
2. Computer Networks and Internets, 5th Ed. — Douglas E. Comer, Prentice Hall, 2008
3. Data and Computer Communications, 8th Ed. — William Stallings, Macmillan, 2006



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## 1. Introduction to Data Communication & Networking

A computer network is a collection of interconnected devices (computers, servers, routers, phones, sensors) that can exchange data and share resources such as files, printers, and internet access. Data communication is the exchange of data between two devices via a transmission medium such as a wire cable.

### 1.1 Components of Data Communication

- Message — the information to be communicated (text, number, image, audio, video).
- Sender — the device that sends the message.
- Receiver — the device that receives the message.
- Medium — the physical path through which the message travels (cable, air/wireless).
- Protocol — a set of rules that governs data communication.

### 1.2 Effectiveness of a Data Communication System

- Delivery — data must be delivered to the correct destination only.
- Accuracy — data must be delivered accurately, without alteration.
- Timeliness — data must be delivered in a timely manner (real-time delivery for audio/video).
- Jitter — variation in packet arrival time must be minimal.

### 1.3 Types of Networks (by Size / Scope)

| Network Type | Full Form                 | Coverage                         | Example                      |
|--------------|---------------------------|----------------------------------|------------------------------|
| PAN          | Personal Area Network     | A few metres (around one person) | Bluetooth headset to phone   |
| LAN          | Local Area Network        | A building or campus             | Office network, computer lab |
| MAN          | Metropolitan Area Network | A city                           | Cable TV network             |
| WAN          | Wide Area Network         | Country / Globe                  | The Internet                 |

### 1.4 Network Topologies

Topology describes how devices (nodes) are physically or logically arranged and connected.



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| Topology | Description                                                    | Advantage                                          | Disadvantage                                         |
|----------|----------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------|
| Bus      | All devices share one common backbone cable                    | Cheap, easy to install                             | Whole network fails if backbone breaks               |
| Star     | All devices connect to a central hub/switch                    | Easy to manage, one failure does not affect others | Central device is a single point of failure          |
| Ring     | Each device connects to exactly two neighbours, forming a ring | Equal access, no collisions with token passing     | Break in ring can disrupt network (unless dual ring) |
| Mesh     | Every device connects to every other device                    | Highly reliable, no traffic bottleneck             | Very expensive, complex cabling                      |
| Hybrid   | Combination of two or more topologies                          | Flexible, scalable                                 | Complex design and management                        |

## 1.5 Network Models (LAN Architectures)

- Client-Server Model — Clients request services; a dedicated server provides them (e.g., web server, file server). Centralized management, better security.
- Peer-to-Peer (P2P) Model — Every computer acts as both client and server; no dedicated server. Simple and cheap but harder to secure/manage at scale.

## 2. Analog and Digital Transmission

### Definition: Analog Signal

A continuous signal that varies smoothly over time and can take infinite values within a range (e.g., sound waves, sine wave).

### Definition: Digital Signal

A discrete signal that has only a limited (finite) number of defined values, typically represented as 0s and 1s (bits).

## 2.1 Analog vs Digital Transmission — Comparison

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| Feature         | Analog Transmission                     | Digital Transmission                     |
|-----------------|-----------------------------------------|------------------------------------------|
| Signal form     | Continuous wave                         | Discrete pulses (0s and 1s)              |
| Noise immunity  | Low — noise adds up and distorts signal | High — signal can be regenerated exactly |
| Bandwidth usage | Less efficient                          | More efficient (compression possible)    |
| Equipment       | Simple, older technology                | Complex, but cheaper due to modern chips |
| Example         | Traditional telephone voice call        | Data sent between computers, digital TV  |

## 2.2 Data Rate Limits

Two key theorems define the maximum rate at which data can be transmitted over a channel:

- Nyquist Theorem (noiseless channel): Bit Rate =  $2 \times B \times \log_2(L)$ , where  $B$  = bandwidth (Hz) and  $L$  = number of signal levels.
- Shannon's Theorem (noisy channel): Capacity  $C = B \times \log_2(1 + S/N)$ , where  $S/N$  is the signal-to-noise ratio.

**Practical / Lab Note:** Numerical Example — Nyquist: If bandwidth  $B = 3000$  Hz and signal levels  $L = 2$  (binary), Bit Rate =  $2 \times 3000 \times \log_2(2) = 2 \times 3000 \times 1 = 6000$  bps.

## 3. Transmission Impairments (Noise)

Signals travelling through a medium always suffer some loss or distortion, collectively called impairments.

| Impairment  | Description                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------|
| Attenuation | Loss of signal strength/energy as it travels through the medium (measured in decibels, dB). Fixed with amplifiers/repeaters. |



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| Impairment       | Description                                                                                                                                        |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Distortion       | Change in the shape of the signal, common in composite signals where each frequency component may travel at a different speed.                     |
| Noise            | Unwanted random signals that get inserted into the data signal during transmission. Types: thermal noise, induced noise, crosstalk, impulse noise. |
| Delay Distortion | Occurs because different frequencies travel at slightly different propagation speeds through a medium.                                             |

## 4. Transmission Media

Transmission media are broadly classified into Guided (wired) and Unguided (wireless).

### 4.1 Guided Media

| Medium             | Description                                                                            | Typical Use                                           |
|--------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|
| Twisted Pair Cable | Two insulated copper wires twisted together to reduce crosstalk (UTP/STP)              | Telephone lines, Ethernet LAN                         |
| Coaxial Cable      | A central copper conductor surrounded by insulation, metallic shield, and outer jacket | Cable TV, older Ethernet (10BASE2)                    |
| Fiber Optic Cable  | Data transmitted as light pulses through glass/plastic fibre core                      | High-speed backbone, long distance, internet backbone |

### 4.2 Unguided (Wireless) Media

- Radio Waves — omnidirectional, used in FM radio, cordless phones, Wi-Fi.
- Microwaves — line-of-sight, unidirectional, used in satellite and cellular communication.
- Infrared — short range, cannot penetrate walls, used in remote controls.



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## 5. Encoding Techniques

Encoding is the process of converting data into signals for transmission over a medium.

| Conversion         | Meaning                                    | Example Technique                             |
|--------------------|--------------------------------------------|-----------------------------------------------|
| Digital-to-Digital | Digital data represented by digital signal | NRZ, Manchester, Differential Manchester      |
| Analog-to-Digital  | Analog data converted to digital signal    | Pulse Code Modulation (PCM), Delta Modulation |
| Digital-to-Analog  | Digital data sent using analog signal      | ASK, FSK, PSK, QAM                            |
| Analog-to-Analog   | Analog data modulated onto analog carrier  | AM, FM, PM                                    |

### 5.1 Common Digital-to-Digital Line Coding Schemes

- NRZ (Non-Return-to-Zero) — Level represents bit value directly; simple but has synchronization problems for long runs of same bit.
- Manchester Encoding — A transition occurs at the middle of every bit period (used in classic Ethernet); provides self-clocking.
- Differential Manchester — Transition at mid-bit for clocking, plus a transition at start of bit for '0', no transition at start for '1'.

## 6. Synchronous vs Asynchronous Transmission

| Feature  | Asynchronous Transmission                                 | Synchronous Transmission                              |
|----------|-----------------------------------------------------------|-------------------------------------------------------|
| Timing   | No shared clock; each character sent independently        | Sender and receiver share a clock signal              |
| Framing  | Start bit + data bits + stop bit(s) around each character | Data sent in continuous blocks/frames with sync bytes |
| Overhead | Higher (extra bits per character)                         | Lower (overhead only per frame)                       |
| Speed    | Slower, simpler, cheaper hardware                         | Faster, used for bulk data transfer                   |



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| Feature | Asynchronous Transmission               | Synchronous Transmission             |
|---------|-----------------------------------------|--------------------------------------|
| Example | Keyboard to computer (old serial ports) | Network transmission between routers |

## 7. Protocol Design Issues

When designing a network protocol, several fundamental issues must be addressed:

- Addressing — how sender and receiver are identified uniquely (IP address, MAC address).
- Error control — detecting and correcting errors introduced during transmission.
- Flow control — matching the transmission rate of a fast sender to a slow receiver.
- Sequencing — ensuring packets are reassembled in the correct order at the destination.
- Connection establishment/release — how a session is opened and closed (e.g., TCP 3-way handshake).
- Multiplexing/Demultiplexing — sharing one physical link among multiple logical connections.
- Routing — choosing the best path from source to destination across multiple networks.



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## 8. Network Architectures: OSI and TCP/IP Models

A network model organizes the communication process into layers, where each layer performs a specific function and provides services to the layer above it.

### 8.1 OSI Reference Model (7 Layers)

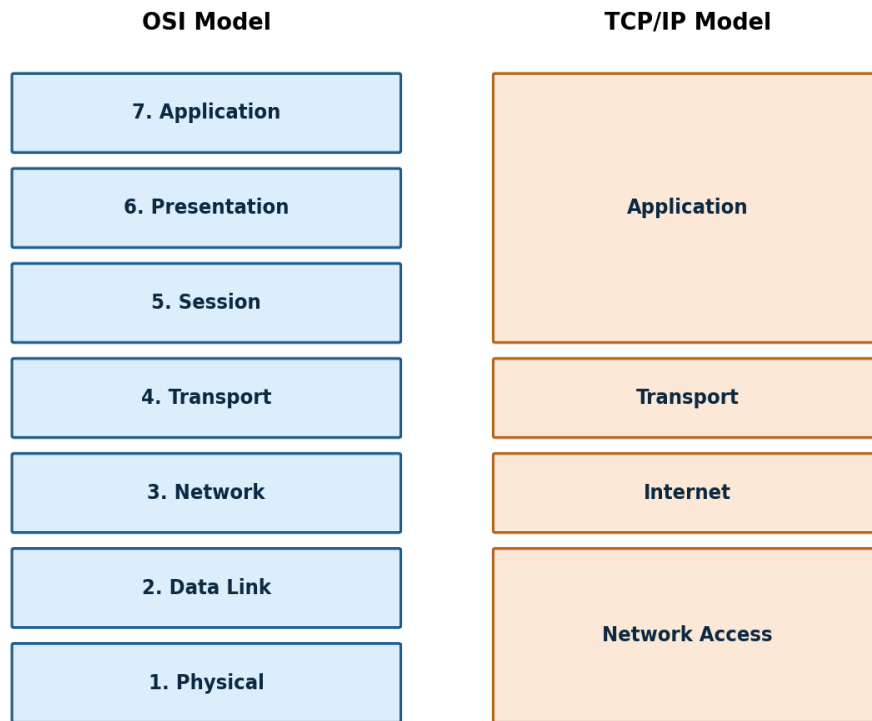


Figure: OSI 7-layer model compared with the TCP/IP 4-layer model

| Layer | Name         | Function                                                            | PDU / Example        |
|-------|--------------|---------------------------------------------------------------------|----------------------|
| 7     | Application  | Interface to end-user applications (email, browsing, file transfer) | HTTP, FTP, SMTP, DNS |
| 6     | Presentation | Data translation, encryption/decryption, compression                | SSL/TLS, JPEG, ASCII |
| 5     | Session      | Establishes, manages, and terminates sessions between applications  | NetBIOS, RPC         |
| 4     | Transport    | End-to-end delivery, segmentation, flow & error control             | TCP, UDP (Segment)   |



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| Layer | Name      | Function                                                      | PDU / Example                      |
|-------|-----------|---------------------------------------------------------------|------------------------------------|
| 3     | Network   | Logical addressing and routing between networks               | IP, ICMP (Packet)                  |
| 2     | Data Link | Physical addressing (MAC), framing, error detection on a link | Ethernet, HDLC, PPP (Frame)        |
| 1     | Physical  | Transmission of raw bits over the physical medium             | Cables, connectors, voltages (Bit) |

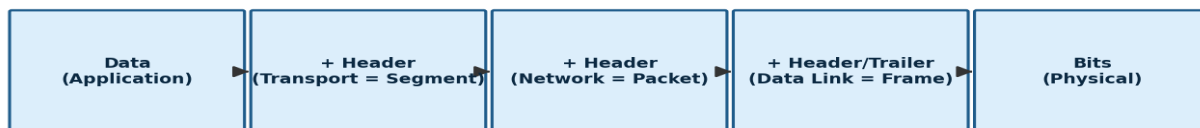
## 8.2 TCP/IP Model (4 Layers)

| Layer                 | Corresponds to OSI                   | Key Protocols                |
|-----------------------|--------------------------------------|------------------------------|
| Application           | Application + Presentation + Session | HTTP, FTP, SMTP, DNS, Telnet |
| Transport             | Transport                            | TCP, UDP                     |
| Internet              | Network                              | IP, ICMP, ARP, RARP          |
| Network Access (Link) | Data Link + Physical                 | Ethernet, Wi-Fi, PPP         |

## 8.3 Encapsulation and Decapsulation (How Data Actually Travels)

Whenever data moves down the protocol stack at the sender, each layer adds its own header (and sometimes a trailer) to the data received from the layer above. This process is called encapsulation. At the receiver, each layer removes (strips) the corresponding header as data moves up the stack — this is decapsulation.

### ENCAPSULATION (Sender side, top to bottom)



At the receiver, the reverse process (DECAPSULATION) removes each header/trailer as data moves bottom to top, until original data is delivered to the application.



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*Figure: Encapsulation process: Data -> Segment -> Packet -> Frame -> Bits*

Step-by-step process (sending an email, as an example):

1. Application layer creates the data (e.g., email text) and hands it to the Transport layer.
2. Transport layer (TCP) adds a header containing source/destination port numbers, sequence number → forms a Segment.
3. Network layer (IP) adds a header with source/destination IP addresses → forms a Packet (datagram).
4. Data Link layer adds a header (source/destination MAC address) and a trailer (CRC for error detection) → forms a Frame.
5. Physical layer converts the frame into a stream of bits (electrical signals, light pulses, or radio waves) and transmits it over the medium.
6. At the receiver, the reverse happens layer by layer (decapsulation) — Physical → Data Link → Network → Transport → Application — until the original data reaches the receiving application.

**Practical / Lab Note:** Exam tip: Remember the PDU (Protocol Data Unit) name changes at every layer: Data (App) → Segment (Transport) → Packet (Network) → Frame (Data Link) → Bits (Physical).



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## 9. Error Control (Detection & Correction)

Errors occur when bits are altered during transmission due to noise. Error control includes error detection (knowing an error occurred) and error correction (fixing it).

### 9.1 Parity Check

A single parity bit is added to make the total number of 1s either even (even parity) or odd (odd parity).

**Practical / Lab Note:** Example: Data = 1011001 (four 1s). For even parity, parity bit = 0 (already even) → transmitted: 10110010. If one bit flips during transmission, the receiver recounts 1s; an odd count reveals an error. Limitation: cannot detect an even number of bit errors.

### 9.2 Checksum

The sender divides data into equal-size segments, adds them using 1's complement arithmetic, and sends the complement of the sum (checksum) along with data. The receiver adds all segments plus the checksum; if the result is all 1s, data is accepted as error-free.

### 9.3 Cyclic Redundancy Check (CRC) — Solved Example

CRC treats the data bits as a polynomial and divides it (using binary/modulo-2 division, i.e., XOR) by a fixed generator polynomial. The remainder is appended to the data as the CRC field.

Example: Data = 1101011011, Generator = 10011 (4-bit CRC, so append 4 zeros).

Dividend (Data + 0000): 11010110110000  
Generator: 10011

Step 1: 1101 0110 110000  
      1001 1  
      -----  
      0100 1  
      ... (continue XOR division bit by bit)

Final Remainder (CRC) = 1110 (example result)  
Transmitted data = Original Data + CRC = 1101011011 1110

At the receiver: the received data (data + CRC) is divided again by the same generator. If the remainder is zero, data is accepted as error-free; a non-zero remainder indicates an error.



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**Practical / Lab Note:** Practical: CRC is implemented in hardware/software for Ethernet frames (FCS field) — students should practice the XOR long-division method by hand for at least 2-3 different generator polynomials (e.g., CRC-4, CRC-8) as it is a common exam/lab question.

## 9.4 Hamming Code — Solved Example (Error Correction)

Hamming code can both detect and correct single-bit errors by inserting redundant parity bits at positions that are powers of 2 (1, 2, 4, 8, ...).

Example: Send data bits 1011 using Hamming Code.

1. Number of parity bits  $r$  must satisfy  $2^r \geq m + r + 1$ . For  $m = 4$  data bits,  $r = 3$  (positions 1, 2, 4).
2. Arrange bits in positions 1-7: P1 P2 D1 P4 D2 D3 D4 = \_ \_ 1 \_ 0 1 1
3. P1 (checks positions 1,3,5,7 = ?,1,0,1)  $\rightarrow P1 = \text{XOR}(1,0,1) = 0$
4. P2 (checks positions 2,3,6,7 = ?,1,1,1)  $\rightarrow P2 = \text{XOR}(1,1,1) = 1$
5. P4 (checks positions 4,5,6,7 = ?,0,1,1)  $\rightarrow P4 = \text{XOR}(0,1,1) = 0$
6. Final transmitted code: 0 1 1 0 0 1 1 (P1 P2 D1 P4 D2 D3 D4)

At the receiver, the same parity groups are recalculated. If all are zero, no error. If non-zero, the binary value formed by the parity results directly points to the exact bit position that is in error, which is then simply flipped to correct it.



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## 10. Flow Control

Flow control ensures a fast sender does not overwhelm a slow receiver by regulating the amount/rate of data sent.

### 10.1 Stop-and-Wait Protocol

Sender transmits one frame and waits for an acknowledgement (ACK) before sending the next frame.

- Simple to implement but very inefficient — link remains idle while waiting for ACK.
- Efficiency = Transmission time / (Transmission time + 2 × Propagation time).

**Practical / Lab Note:** Numerical Example: If frame transmission time = 1 ms and one-way propagation delay = 5 ms, then Efficiency =  $1 / (1 + 2 \times 5) = 1/11 \approx 9\%$ . This shows how poor stop-and-wait becomes over long-delay links (e.g., satellite links).

### 10.2 Sliding Window Protocol

Allows the sender to transmit multiple frames (up to a 'window size') before requiring an acknowledgement, improving link utilization.

#### Sliding Window (Go-Back-N, window size = 4)



Sender can transmit all frames within the window.  
On receiving ACK for F0, window slides forward by one.  
If a frame is lost/errored, receiver discards subsequent frames and sender must retransmit from the lost frame onward.

Figure: Sliding window mechanism (window size = 4)

#### 10.2.1 Go-Back-N ARQ

- Sender can send N frames without waiting for individual ACKs (N = window size).
- Receiver only accepts frames in correct order; if a frame is lost/errored, it discards all subsequent frames.
- Sender must retransmit the lost frame AND all frames sent after it — less efficient on lossy links but simple receiver logic.



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## 10.2.2 Selective Repeat ARQ

- Receiver individually buffers and acknowledges each correctly received frame, even if out of order.
- Only the specific lost/errored frame is retransmitted — more efficient but requires more complex buffering at the receiver.

**Practical / Lab Note:** Numerical Example: For a link with bandwidth 1 Mbps and round-trip time (RTT) = 20 ms, if frame size = 1000 bits, then to keep the pipe full the required window size  $W \geq (\text{Bandwidth} \times \text{RTT}) / \text{Frame size} = (1,000,000 \times 0.02) / 1000 = 20$  frames.

## 11. Data Link Layer Protocols: HDLC & PPP

### 11.1 HDLC (High-Level Data Link Control)

HDLC is a bit-oriented protocol for point-to-point and multipoint links, supporting three frame types: Information (I), Supervisory (S), and Unnumbered (U) frames.

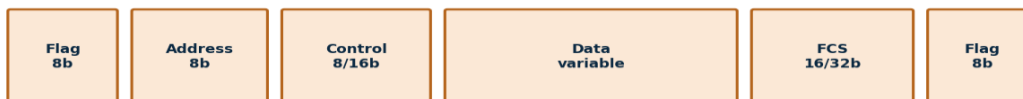


Figure: HDLC frame format

- Flag — 8-bit pattern (01111110) marking start/end of the frame.
- Address — identifies the secondary station (in multipoint configuration).
- Control — identifies frame type and carries sequence numbers for flow/error control.
- FCS (Frame Check Sequence) — CRC used for error detection.

### 11.2 PPP (Point-to-Point Protocol)

PPP is widely used for direct connections between two nodes, such as dial-up/DSL internet connections. It supports authentication (PAP, CHAP), multiple network layer protocols, and error detection but not flow control or error correction.

- Uses LCP (Link Control Protocol) to establish, configure, and test the link.
- Uses NCP (Network Control Protocol) to negotiate network-layer options (e.g., IPCP for IP).



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## 12. Local Area Networks (LAN) & MAC Layer

### 12.1 Ethernet

Ethernet is the most widely used LAN technology, standardized as IEEE 802.3. It originally used CSMA/CD as its medium access control method (in shared/bus topology); modern switched Ethernet uses full-duplex links, eliminating collisions.



Figure: Ethernet frame format

#### 12.1.1 CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

1. Carrier Sense — station listens to the channel before transmitting.
2. Multiple Access — many stations share the same medium.
3. If channel is idle, station transmits; if busy, it waits.
4. Collision Detection — while transmitting, the station keeps listening; if a collision is detected, it stops immediately, sends a jam signal, and waits a random back-off time (binary exponential backoff) before retrying.

### 12.2 Token Ring

Token Ring (IEEE 802.5) uses a special frame called a token that circulates around the ring. Only the station holding the token may transmit data, which eliminates collisions entirely (deterministic access, unlike Ethernet's probabilistic CSMA/CD).

- Advantage: fair, predictable access; performs well under heavy load.
- Disadvantage: token loss or a broken ring can halt the entire network (mitigated using dual-ring / active monitor mechanisms).

## 13. Multiplexing

Multiplexing combines multiple signals from different sources into one signal over a shared medium, to make efficient use of bandwidth.



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| Type            | Full Form               |          | Description                                                                                       | Example                       |
|-----------------|-------------------------|----------|---------------------------------------------------------------------------------------------------|-------------------------------|
| FDM             | Frequency Multiplexing  | Division | Each signal is assigned a different frequency band within the shared bandwidth (analog technique) | Cable TV, radio broadcasting  |
| TDM             | Time Multiplexing       | Division | Each signal is assigned a fixed time slot in rotation (digital technique)                         | Digital telephone (T1 line)   |
| WDM             | Wavelength Multiplexing | Division | Multiple light wavelengths carried on a single fibre-optic cable                                  | Fibre-optic backbone networks |
| Statistical TDM | —                       |          | Time slots are dynamically allocated only to active senders (not fixed), improving efficiency     | Packet-based networks         |

## 14. Switching

Switching is the technique of forwarding data through intermediate nodes from source to destination across a network.

| Type                               | Description                                                                                                                      |  | Example                                         |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------|
| Circuit Switching                  | A dedicated physical path/circuit is established for the entire duration of the communication before data transfer begins.       |  | Traditional telephone (PSTN) calls              |
| Packet Switching (Datagram)        | Data is broken into packets; each packet is routed independently and may take different paths; reassembled at destination.       |  | The Internet (IP)                               |
| Packet Switching (Virtual Circuit) | A logical path is set up before data transfer, and all packets follow the same path, though not physically dedicated.            |  | ATM, Frame Relay                                |
| Message Switching                  | The entire message is sent to an intermediate switch, stored, and then forwarded (store-and-forward) when the next link is free. |  | Old telegraph networks, email store-and-forward |



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## 15. IP Networks & IP Addressing

### 15.1 IPv4 Address Structure

An IPv4 address is a 32-bit number, usually written in dotted-decimal notation (e.g., 192.168.1.1), divided into a Network ID and a Host ID.

### 15.2 Classful Addressing

| Class | First Octet Range | Default Mask        | Usable Hosts per Network | Typical Use                |
|-------|-------------------|---------------------|--------------------------|----------------------------|
| A     | 1 – 126           | 255.0.0.0 (/8)      | 16,777,214               | Very large networks        |
| B     | 128 – 191         | 255.255.0.0 (/16)   | 65,534                   | Medium/large organizations |
| C     | 192 – 223         | 255.255.255.0 (/24) | 254                      | Small networks             |
| D     | 224 – 239         | —                   | —                        | Multicast                  |
| E     | 240 – 255         | —                   | —                        | Reserved / experimental    |

### 15.3 Subnetting — Solved Numerical Examples

Subnetting divides a large network into smaller sub-networks (subnets) to improve address utilization, security, and manageability.

**Example 1: Given network 192.168.1.0/24, create 4 subnets.**

1. We need 4 subnets  $\rightarrow$  need 2 extra bits ( $2^2 = 4$ ), since borrowing 2 bits from the host portion gives 4 subnets.
2. New subnet mask =  $/24 + 2 = /26 \rightarrow 255.255.255.192$
3. Block size =  $256 - 192 = 64$
4. Subnet 1: 192.168.1.0 – 192.168.1.63 (usable hosts: .1 to .62)
5. Subnet 2: 192.168.1.64 – 192.168.1.127 (usable hosts: .65 to .126)
6. Subnet 3: 192.168.1.128 – 192.168.1.191 (usable hosts: .129 to .190)
7. Subnet 4: 192.168.1.192 – 192.168.1.255 (usable hosts: .193 to .254)



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### Example 2: How many hosts can a /28 network support?

1. /28 means 28 network bits, so  $32 - 28 = 4$  host bits remain.
2. Total addresses =  $2^4 = 16$
3. Usable hosts =  $2^4 - 2$  (subtracting network address and broadcast address) = 14 hosts.

### Example 3: Find the network address for IP 172.16.45.100 with mask 255.255.240.0 (/20).

1. Convert third octet 45 to binary: 00101101
2. Mask third octet 240 = 11110000 — perform bitwise AND: 00101101 AND 11110000 = 00100000 = 32
3. Network address = 172.16.32.0

**Practical / Lab Note:** Practical Lab Tip: Practice subnetting using the 'magic number' method (256 – subnet mask octet value = block size) — this is the fastest technique for exams and interviews, and can be verified using tools like ipcalc or an online subnet calculator.

## 15.4 IPv6 (Brief Overview)

IPv6 uses 128-bit addresses (written in 8 groups of 4 hexadecimal digits, e.g., 2001:0db8:85a3::8a2e:0370:7334) to solve IPv4 address exhaustion, and includes built-in support for security (IPsec) and simplified header format.

## 16. Routing & Bridging

### 16.1 Routing

Routing is the process by which a router selects the best path to forward a packet toward its destination network.

| Type            | Description                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Static Routing  | Routes are manually configured by the network administrator; does not adapt automatically to network changes. |
| Dynamic Routing | Routers automatically exchange information and adjust routes using routing protocols.                         |

#### 16.1.1 Distance Vector Routing



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Each router maintains a table of distances (usually hop count) to every other network and periodically shares this table with its direct neighbours (e.g., RIP – Routing Information Protocol, max 15 hops). Uses the Bellman-Ford algorithm.

### 16.1.2 Link State Routing

Each router builds a complete map (topology) of the entire network by flooding link-state information to all routers, then independently computes the shortest path using Dijkstra's algorithm (e.g., OSPF – Open Shortest Path First).

### 16.1.3 Path Vector Routing

Used between autonomous systems on the Internet (e.g., BGP – Border Gateway Protocol); routing decisions are based on policies and the full path (list of AS numbers) rather than just distance.

### 16.2 Bridging

A bridge operates at the Data Link layer and connects two LAN segments, forwarding frames based on MAC addresses while filtering out unnecessary traffic, thereby reducing collisions and segmenting collision domains. Modern switches are essentially multiport bridges.



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## 17. Transport Layer: TCP & UDP

### 17.1 TCP (Transmission Control Protocol)

TCP is a connection-oriented, reliable transport protocol that guarantees ordered, error-checked delivery of a stream of data between applications.

- Connection-oriented — a connection must be established (3-way handshake) before data transfer.
- Reliable — uses acknowledgements, retransmission, and sequence numbers.
- Flow control — uses sliding window mechanism.
- Congestion control — adjusts sending rate based on network congestion (e.g., slow start, congestion avoidance).

#### 17.1.1 TCP Three-Way Handshake

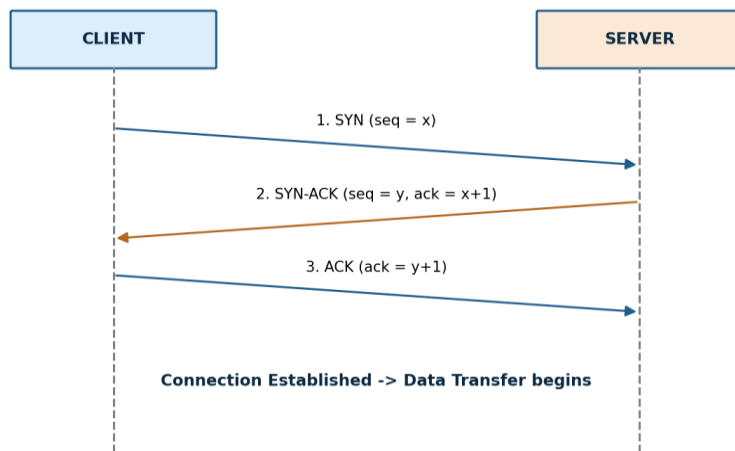


Figure: TCP connection establishment (three-way handshake)

1. Client sends a SYN (synchronize) segment with an initial sequence number  $x$ .
2. Server responds with SYN-ACK, acknowledging  $x+1$  and sending its own sequence number  $y$ .
3. Client sends ACK, acknowledging  $y+1$  — connection is now established, and data transfer can begin.

Connection termination uses a similar four-step process involving FIN and ACK segments from both sides.

### 17.2 UDP (User Datagram Protocol)



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UDP is a connectionless, unreliable transport protocol with minimal overhead — it does not guarantee delivery, order, or error recovery, but is fast and simple.

- Used where speed matters more than reliability: DNS queries, video streaming, VoIP, online gaming.
- Header is only 8 bytes (source port, destination port, length, checksum) compared to TCP's minimum 20 bytes.

## 17.3 TCP vs UDP — Comparison

| Feature     | TCP                              | UDP                          |
|-------------|----------------------------------|------------------------------|
| Connection  | Connection-oriented              | Connectionless               |
| Reliability | Reliable (ACK, retransmission)   | Unreliable (best-effort)     |
| Ordering    | Guaranteed in-order delivery     | No ordering guarantee        |
| Speed       | Slower (more overhead)           | Faster (minimal overhead)    |
| Header size | 20-60 bytes                      | 8 bytes                      |
| Use case    | Web (HTTP), Email, File transfer | Streaming, DNS, VoIP, gaming |

## 18. Network Security Issues

As data travels across shared and public networks, it is exposed to various threats. Basic security concepts every networking student must know:

- Confidentiality — ensuring data is only accessible to authorized parties (achieved via encryption, e.g., AES, RSA).
- Integrity — ensuring data is not altered in transit (achieved via hashing, e.g., SHA-256, and digital signatures).
- Availability — ensuring services remain accessible (threatened by Denial-of-Service / DoS attacks).
- Authentication — verifying the identity of communicating parties (passwords, certificates, CHAP/PAP).
- Common threats — eavesdropping/sniffing, spoofing, man-in-the-middle attacks, malware, phishing.



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- Common defenses — firewalls (packet filtering), VPNs (encrypted tunnels), IDS/IPS (intrusion detection/prevention systems), SSL/TLS for secure web traffic.



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## 19. Practical / Lab Work Guide

Since this course includes a weekly lab component, the following hands-on exercises map directly to the theory covered above and are commonly assessed in university labs/vivas:

### 19.1 Networking Commands (any OS terminal)

| Command                       | Purpose                                                                         |
|-------------------------------|---------------------------------------------------------------------------------|
| ping <host>                   | Tests reachability of a host and measures round-trip time                       |
| ipconfig / ifconfig / ip addr | Displays the device's own IP configuration (Windows / Linux-Mac / modern Linux) |
| tracert / traceroute          | Shows the route (hop by hop) a packet takes to reach a destination              |
| netstat -an                   | Displays active network connections and listening ports                         |
| nslookup / dig                | Queries DNS servers to resolve domain names to IP addresses                     |
| arp -a                        | Displays the ARP cache (IP-to-MAC address mappings)                             |

### 19.2 Cisco Packet Tracer / GNS3 Exercises

- Build a simple LAN with a switch and 4 PCs; configure static IP addressing and verify connectivity with ping.
- Connect two LANs through a router; configure subnetted IP addresses on each interface (apply Section 15.3 numericals).
- Configure a simple RIP or OSPF routing protocol between 3 routers and observe route propagation.
- Capture and inspect Ethernet frame headers using the simulator's simulation mode.

### 19.3 Wireshark Exercises

- Capture live traffic and identify the Ethernet, IP, and TCP/UDP headers inside a single packet — relate every field back to Section 8 (Encapsulation).
- Filter for 'tcp.flags.syn==1' to visually observe the three-way handshake described in Section 17.1.1.



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- Capture a DNS query/response and identify it as UDP traffic.

## 19.4 Programming Exercises

- Write a program (C/Python) to compute and verify CRC checksum for a given bit stream (relates to Section 9.3).
- Write a program to encode/decode a message using Hamming Code, and simulate a single-bit error and its correction (Section 9.4).
- Implement a simple Client-Server chat application using TCP sockets, and a UDP version to compare reliability (Section 17).
- Write a subnet calculator that takes an IP address and prefix length and outputs network address, broadcast address, and usable host range.

**Practical / Lab Note:** Viva tip: Examiners frequently ask you to trace through a CRC division, a subnetting problem, or the TCP handshake step-by-step on paper/whiteboard — practice writing these out by hand, not just running code.



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## 20. Solved Numerical Problems (Exam Practice)

### Q1. Bandwidth-Delay Product

A link has a bandwidth of 4 Mbps and a propagation delay of 20 ms. Find the bandwidth-delay product and interpret it.

Bandwidth-Delay Product = Bandwidth x Delay

= 4,000,000 bps x 0.020 s

= 80,000 bits = 10,000 bytes

Interpretation: The link can hold 80,000 bits 'in flight' at any instant (i.e., this is the ideal window size in bits for full utilization).

### Q2. Stop-and-Wait Efficiency

Frame size = 1000 bits, Bandwidth = 1 Mbps, Propagation delay = 15 ms (one-way). Find efficiency.

Transmission time ( $T_t$ ) = Frame size / Bandwidth = 1000 / 1,000,000 = 1 ms

Propagation time ( $T_p$ ) = 15 ms

Efficiency =  $T_t / (T_t + 2T_p) = 1 / (1 + 30) = 1/31 = 3.2\%$

### Q3. Number of Subnets and Hosts

A network 10.0.0.0/8 is subnetted using a /12 mask. Find the number of subnets created and hosts per subnet.

Borrowed bits = 12 - 8 = 4 bits -> Number of subnets =  $2^4 = 16$

Remaining host bits = 32 - 12 = 20 bits

Hosts per subnet =  $2^{20} - 2 = 1,048,574$

### Q4. Simple Checksum Calculation

Sender has two 8-bit data segments: 10101001 and 00111001. Find the checksum to be sent.

Step 1: Add segments (binary addition):

```
  10101001
+ 00111001
-----
 11110010  (9 bits - overflow of 1 bit)
```

Step 2: Wrap around the carry (1's complement addition):

```
  1110010
+         1
-----
```



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11100011 (this is the sum)

Step 3: Checksum = complement of sum = 00011100

This checksum is transmitted along with the data.



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## 21. Quick Revision Summary

| Topic             | One-line Recap                                                                          |
|-------------------|-----------------------------------------------------------------------------------------|
| OSI Model         | 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application   |
| TCP/IP Model      | 4 layers: Network Access, Internet, Transport, Application                              |
| Encapsulation     | Data -> Segment -> Packet -> Frame -> Bits (add header at each layer going down)        |
| Error Detection   | Parity, Checksum, CRC (division-based)                                                  |
| Error Correction  | Hamming Code (parity bits at power-of-2 positions)                                      |
| Flow Control      | Stop-and-Wait (1 frame at a time); Sliding Window (Go-Back-N / Selective Repeat)        |
| Ethernet Access   | CSMA/CD — listen, transmit, detect collision, backoff & retry                           |
| Token Ring Access | Deterministic — only token holder transmits                                             |
| Multiplexing      | FDM (frequency), TDM (time), WDM (wavelength on fibre)                                  |
| Switching         | Circuit (dedicated path), Packet (independent routing), Message (store-and-forward)     |
| Subnetting        | Borrow host bits to create subnets: new hosts = $2^{\text{remaining bits}} - 2$         |
| Routing           | Distance Vector (RIP, Bellman-Ford) vs Link State (OSPF, Dijkstra) vs Path Vector (BGP) |
| TCP               | Connection-oriented, reliable, 3-way handshake (SYN, SYN-ACK, ACK)                      |
| UDP               | Connectionless, unreliable, fast, small 8-byte header                                   |
| Security Pillars  | Confidentiality, Integrity, Availability, Authentication (CIA + A)                      |

*End of Notes — prepared strictly according to the given course outline and the three prescribed reference textbooks (Tanenbaum, Comer, Stallings). Recommended: revise the diagrams and re-derive each solved numerical by hand before the exam.*